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# HTTP Log Analysis Lab Report

## Using Splunk Basics – HTTP Log Analysis Lab Report

**Structured Cybersecurity Report**

## OBJECTIVE OF ANALYSIS

The objective of this analysis is to investigate HTTP traffic logs using Splunk in order to identify suspicious patterns, anomalies, and potential indicators of malicious activity.
This lab simulates a real-world SOC workflow where raw network logs are analyzed to:

During this analysis:

- HTTP logs are ingested into Splunk for processing
- Client-server communication is examined
- Server-side errors and anomalies are identified
- Suspicious user-agents and automated tools are detected
- Large data transfers are analyzed for possible data leakage

This approach helps in developing a basic understanding of log-based threat detection.

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### ANALYSIS TOOLS AND ENVIRONMENT SETUP

The analysis was performed using Splunk, which is a Security Information and Event Management (SIEM) platform used for log monitoring and data analysis.
The dataset consists of HTTP logs generated by Zeek, a network monitoring tool that captures web traffic in structured format.
Key Components:

- Splunk: Used for searching, analyzing, and visualizing log data
- Zeek HTTP Logs: Provides detailed records of web requests and responses
- Log Format (JSON): Enables automatic field extraction in Splunk
- Index (HTTP lab): Used to store and organize HTTP log data
- Source type (Json): Helps Splunk interpret the structure of logs

This environment allows efficient querying and analysis of network activity.

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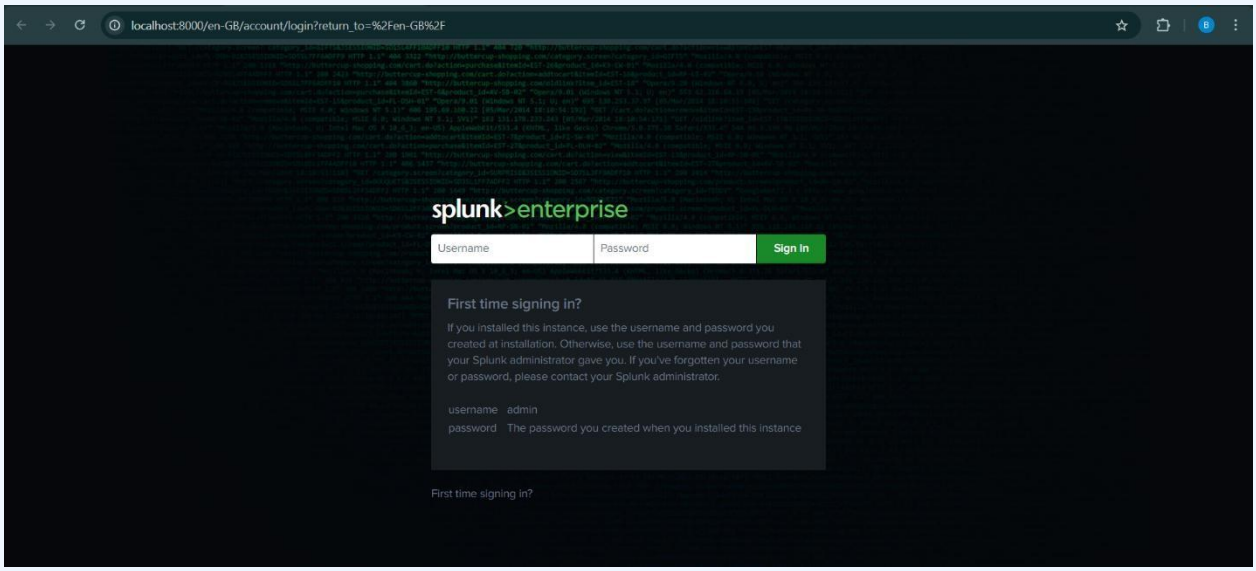
## LAB SETUP

Before starting the analysis, the HTTP log dataset was uploaded into Splunk for processing.

The ingestion process ensures that logs are properly indexed and fields are extracted for analysis.

**Step 1 – Open Splunk Web**

1. Open **Splunk Web Interface** in your browser.
2. Log in using your credentials.

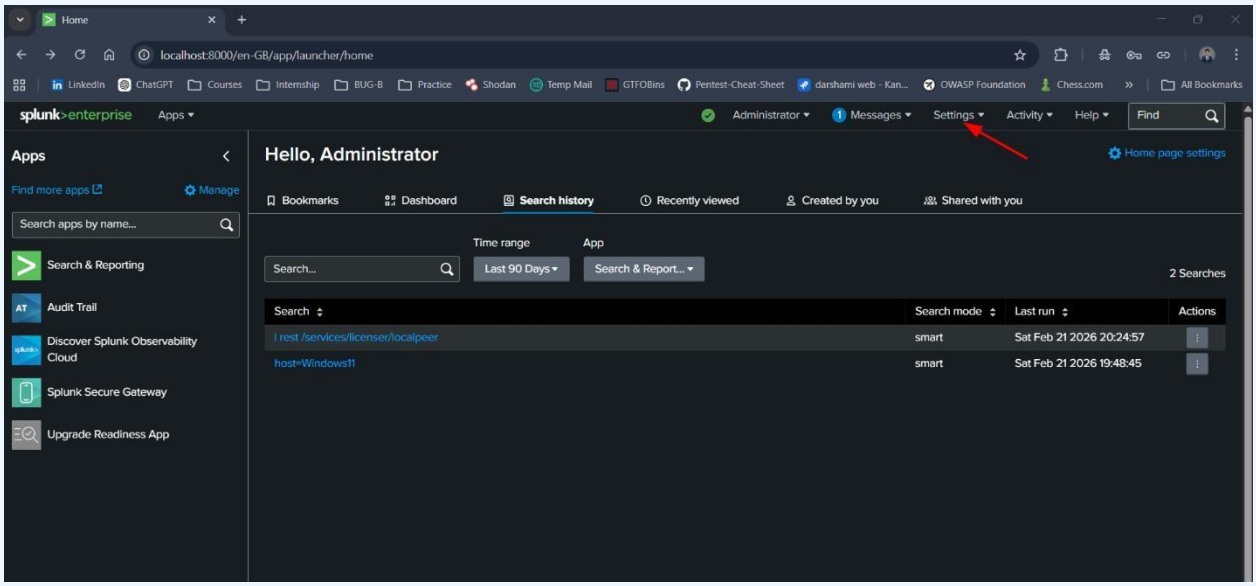


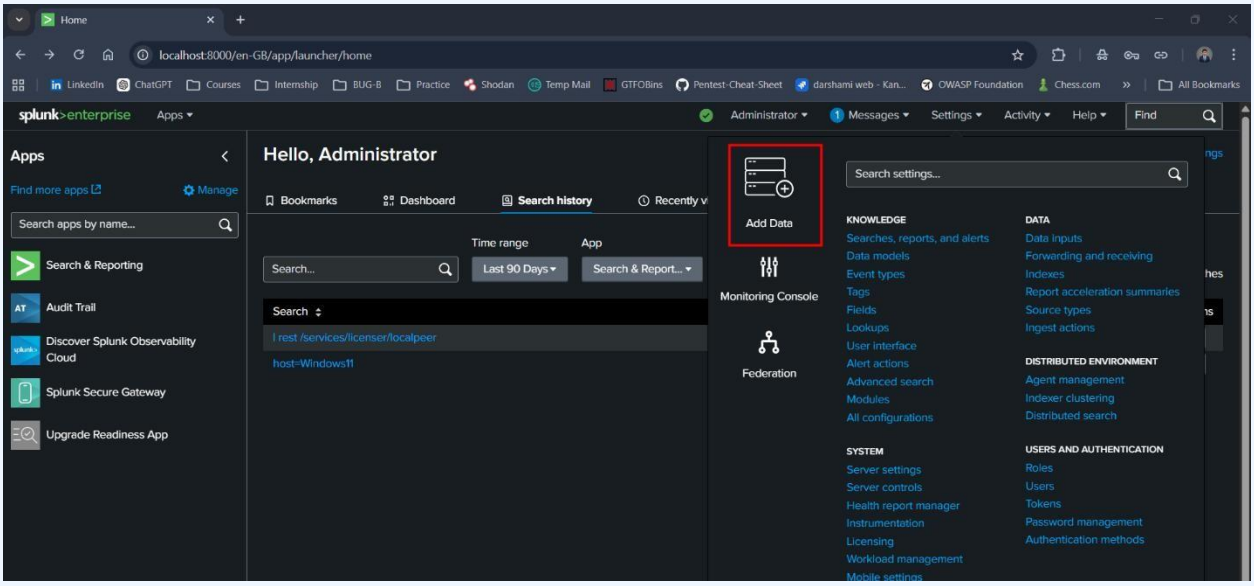

## Step 2 – Navigate to Add Data

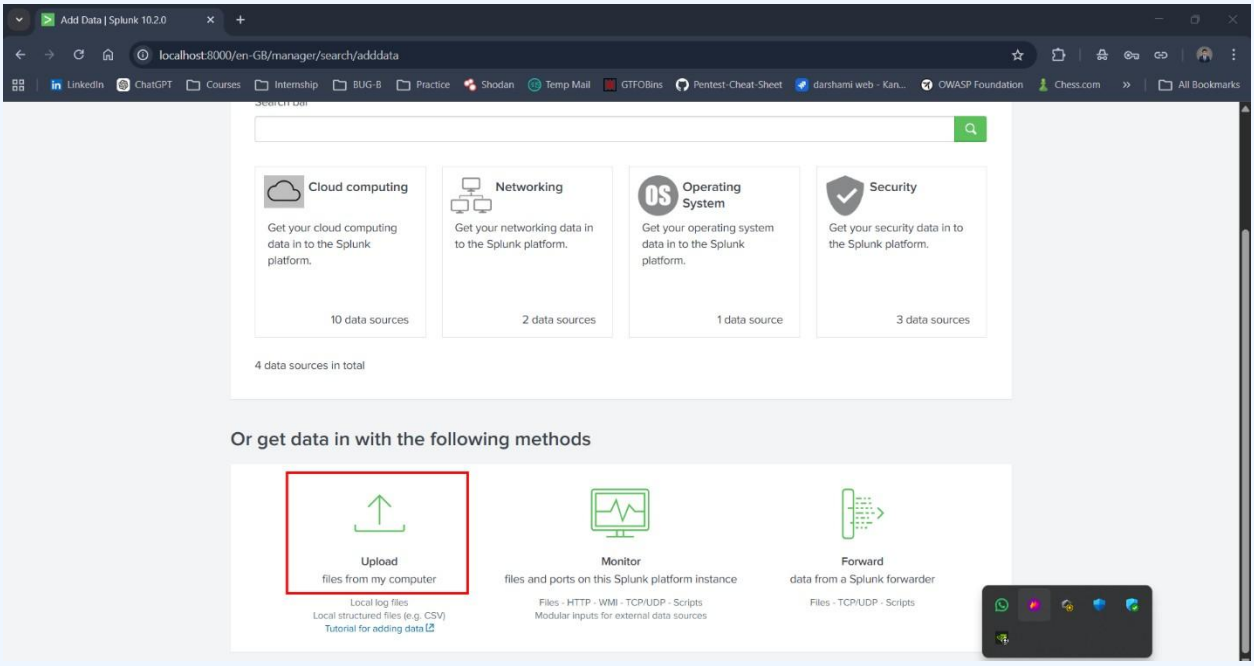
1. Go to:

**Settings → Add Data**

2. Select **Upload.**





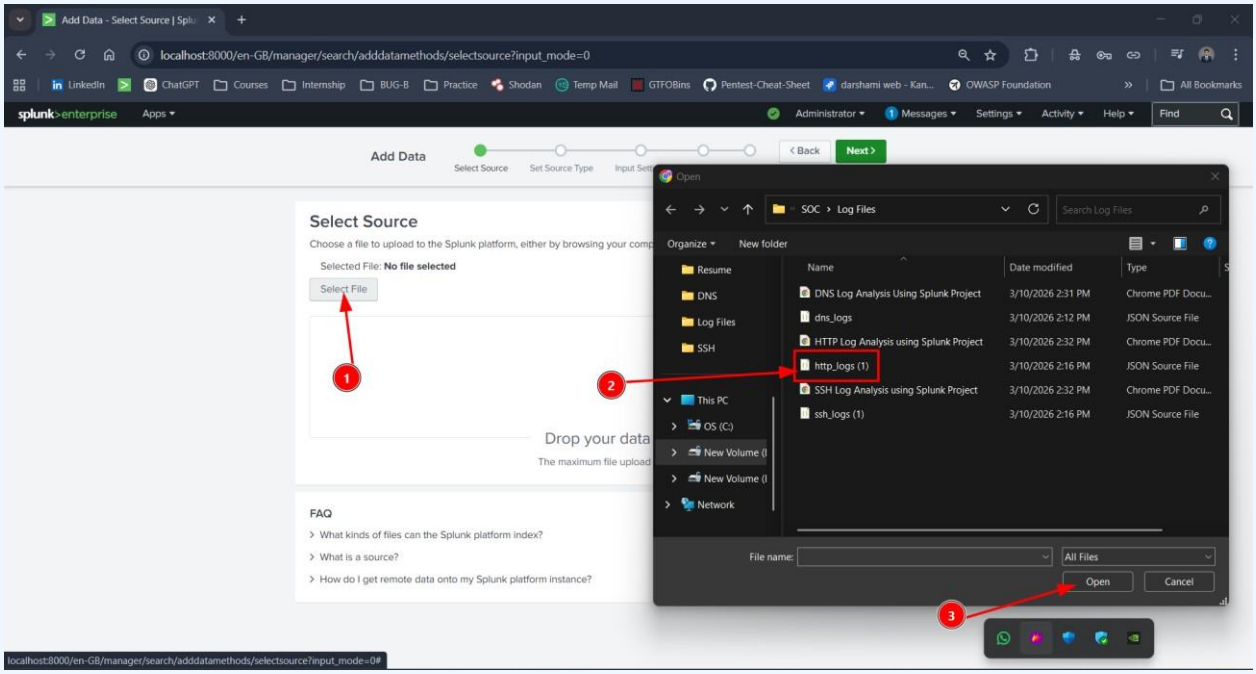

**Step 3 – Upload the HTTP Log File** 1.

Click **Select File**

2. Upload the file:

**http_logs.json**

This dataset contains **HTTP requests and responses recorded by Zeek network monitoring tool**.




**Step 4 – Configure Source Type** Set

the source type to:

**_json**

or create a custom source type:

**http_logs**

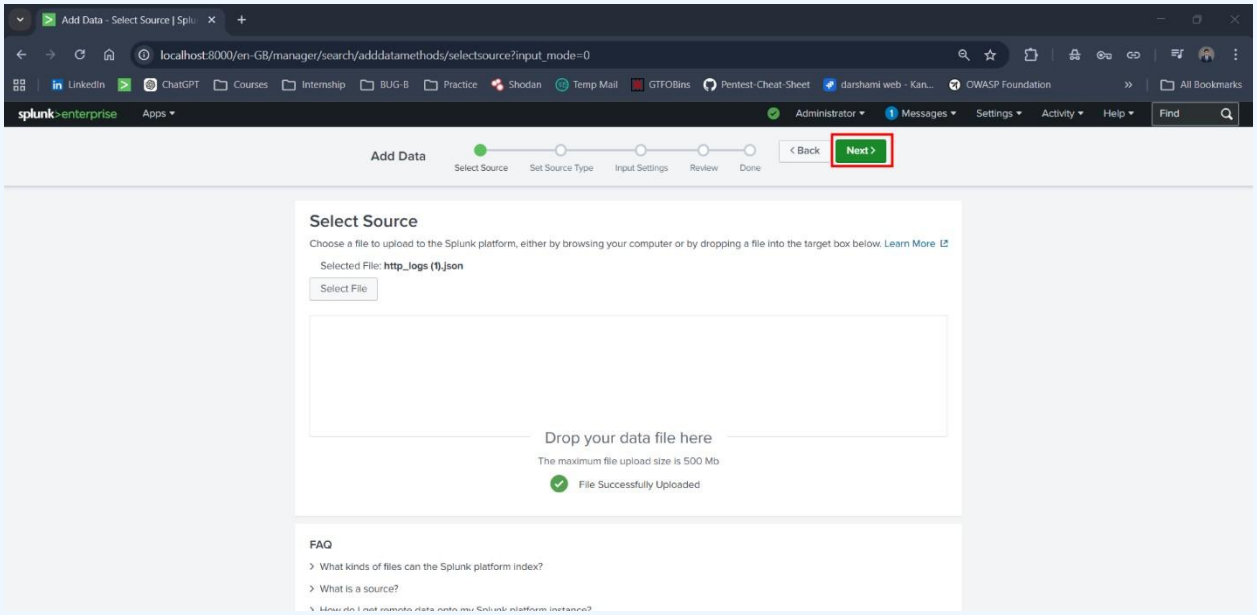
**Reason**

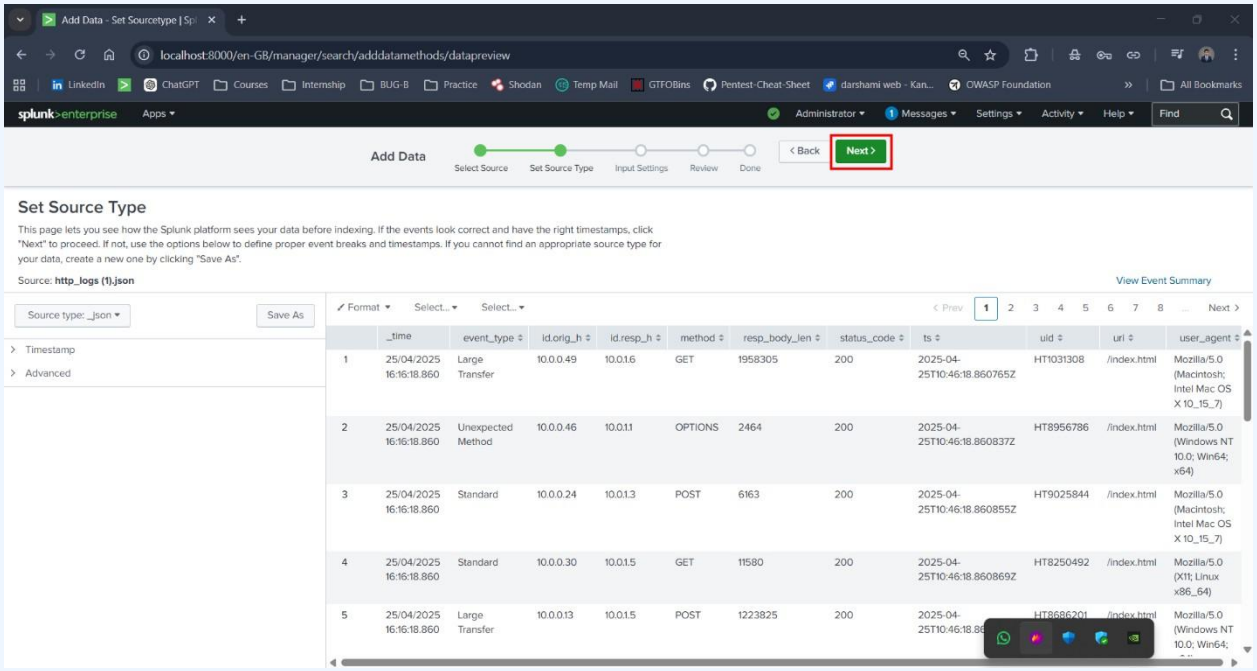
- The log file is in JSON format, so it does not require manual parsing
- Splunk can automatically detect and extract important information from the logs
- This makes the analysis process faster and more accurate

**Important Fields Extracted**

After uploading and setting the source type, the following fields were automatically available:

- id.orig_h – shows the IP address of the client
- id.resp_h – shows the IP address of the server
- uri – shows which page or resource was requested
- user agent – shows the tool or browser used
- status code – shows the response given by the server
- resp_body_len – shows the size of the response




**Step 5 – Configure Index**

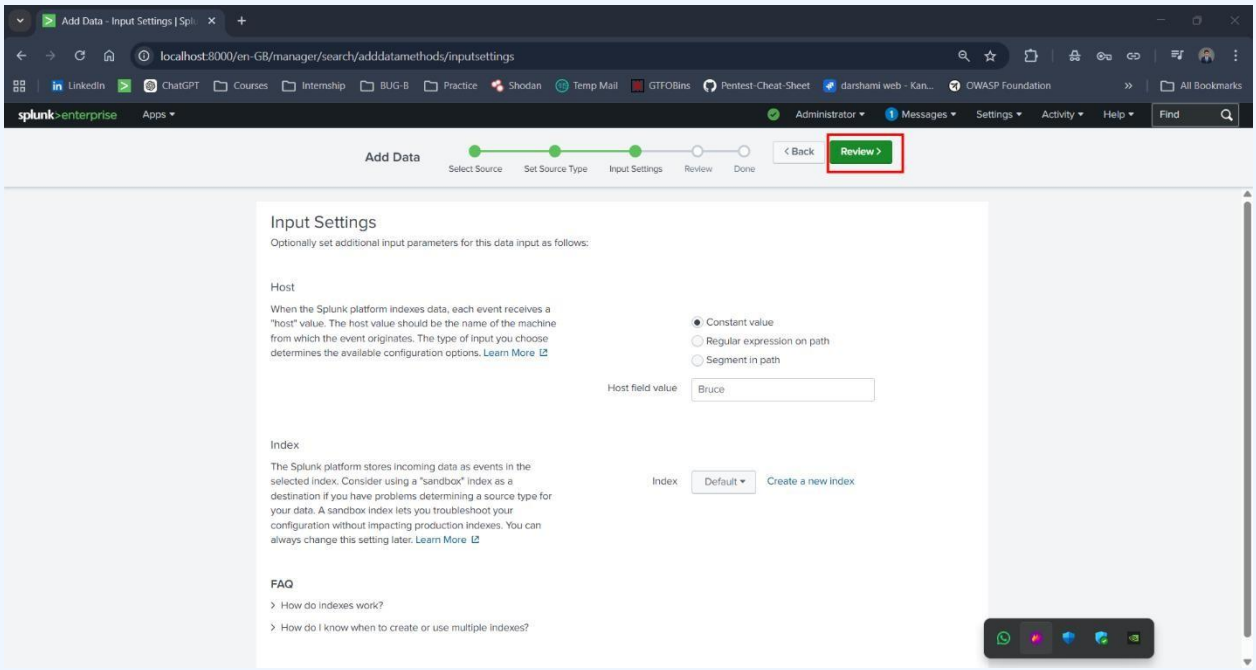
Select or create the index:

**http_lab**

An index named http_lab was selected to store the HTTP log data.

**Reason**

- Using a separate index makes it easier to manage HTTP-related logs.
- It avoids mixing this data with other logs in the system.
- Searching and filtering becomes quicker and more efficient.




**Step** Review the configuration.

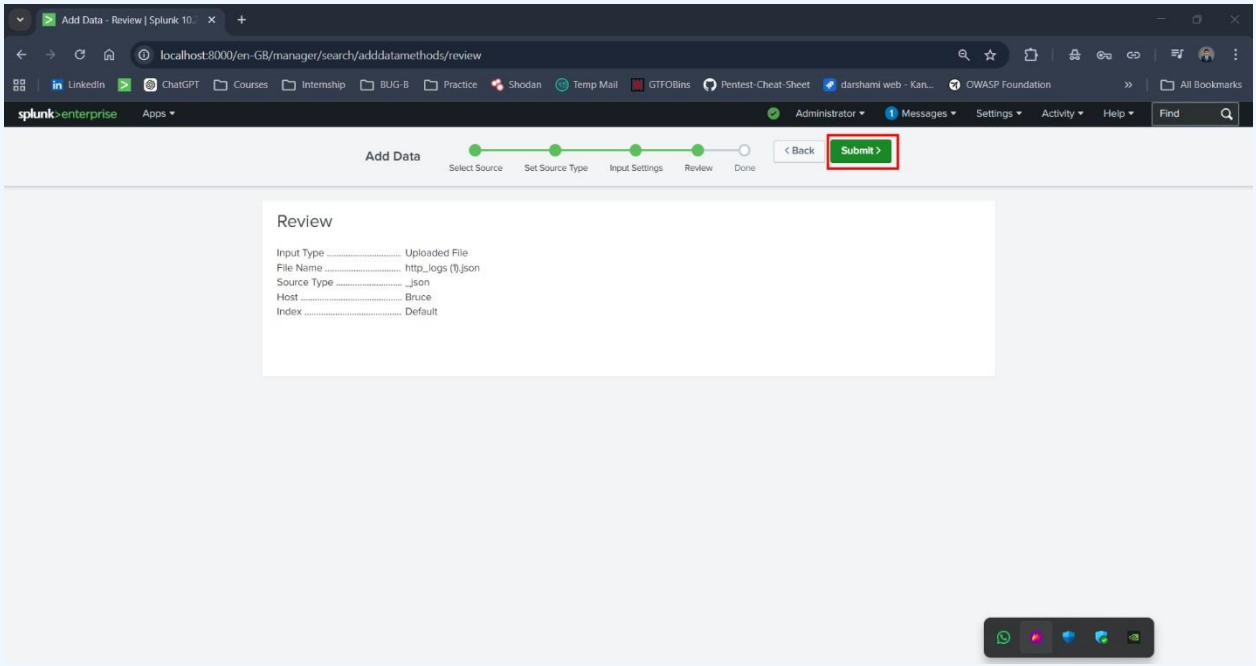
2. Click **Submit.**

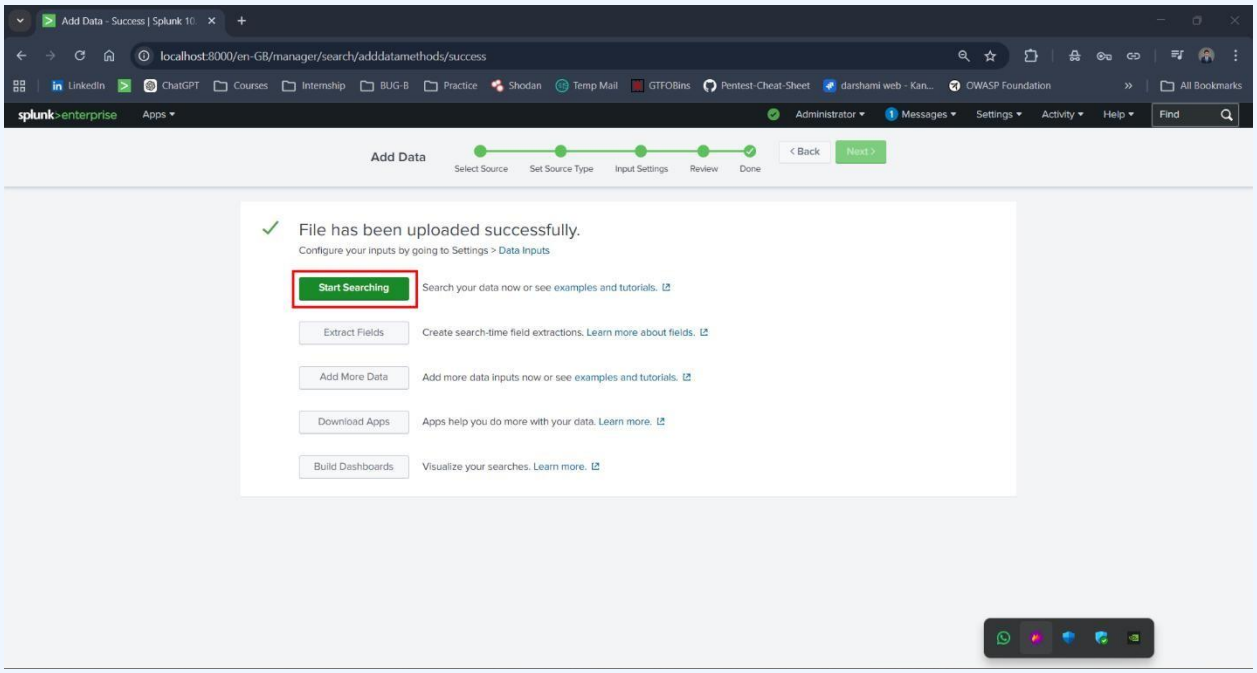
To confirm indexing run:

```
index=http_labs
```

If events appear, the data ingestion was successful.

**6 – Confirm Upload.**



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## UNDERSTANDING HTTP LOGS

HTTP logs record the communication between clients and web servers, including both requests and responses. These logs help in understanding how users interact with a web application.

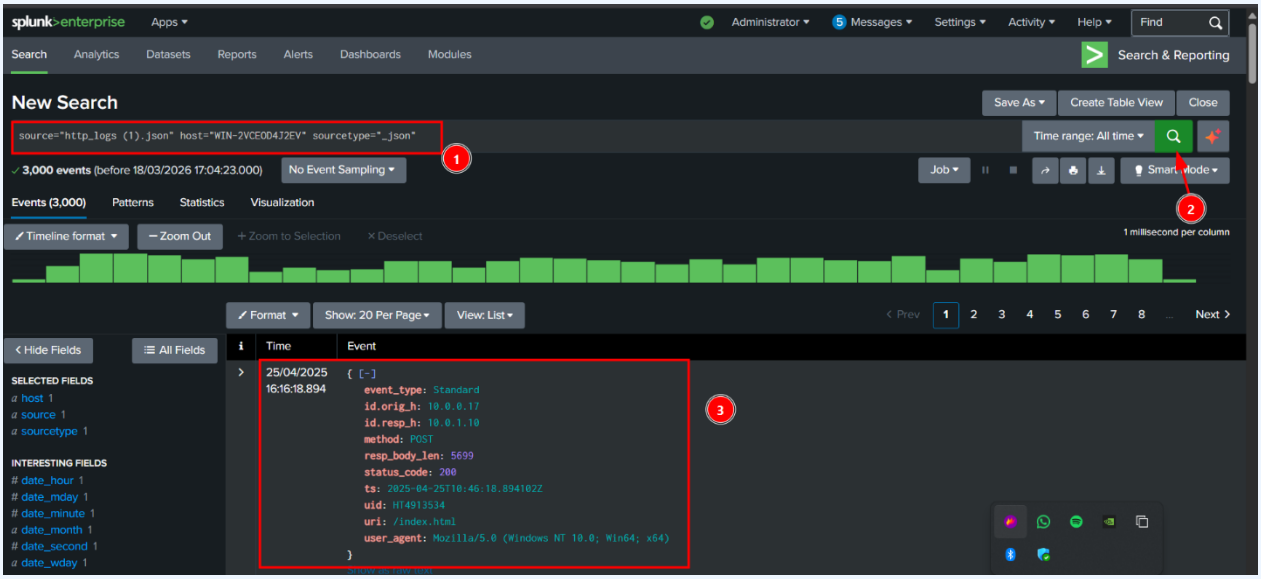
The following fields are commonly found in Zeke HTTP logs:

- **id.orig_h** represents the IP address of the client sending the request.
- **id.resp_h** represents the IP address of the web server receiving the request.
- **uri** shows the specific resource or page requested by the client.
- **user_agent** indicates the application or tool used to make the request.
- **status_code** shows the response returned by the server.
- **resp_body_len** represents the size of the response sent by the server.

These fields are useful for analyzing traffic patterns and identifying unusual or suspicious activity.

Example HTTP Log Event:

id.orig_h: 192.168.1.10
id.resp_h: 192.168.1.5
uri: /login.php
user_agent: curl/7.68.0
status_code: 200
resp_body_len: 10240

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## LAB TASKS

### Task 1: Find Top 10 Endpoints Generating Web Traffic

### SPL Query

Index=http_lab sourcetype=" json"
| status count by "id. orign_h"
| sort -count
| head 10

### Query Explanation

The SPL query used in this step is designed to analyze HTTP traffic and identify the top source IP addresses generating requests.

- **index=http_lab**
  Specifies that the search is performed on the HTTP log data stored in the selected index.
- **sourcetype="json"**
  Filters the data to include only JSON formatted logs.
- **stats count by id.orig_h**
  Counts the number of HTTP requests generated by each client IP address.
- **sort -count**
  Arranges the results in descending order based on request count.
- **head 10**
  Displays the top 10 IP addresses with the highest traffic.

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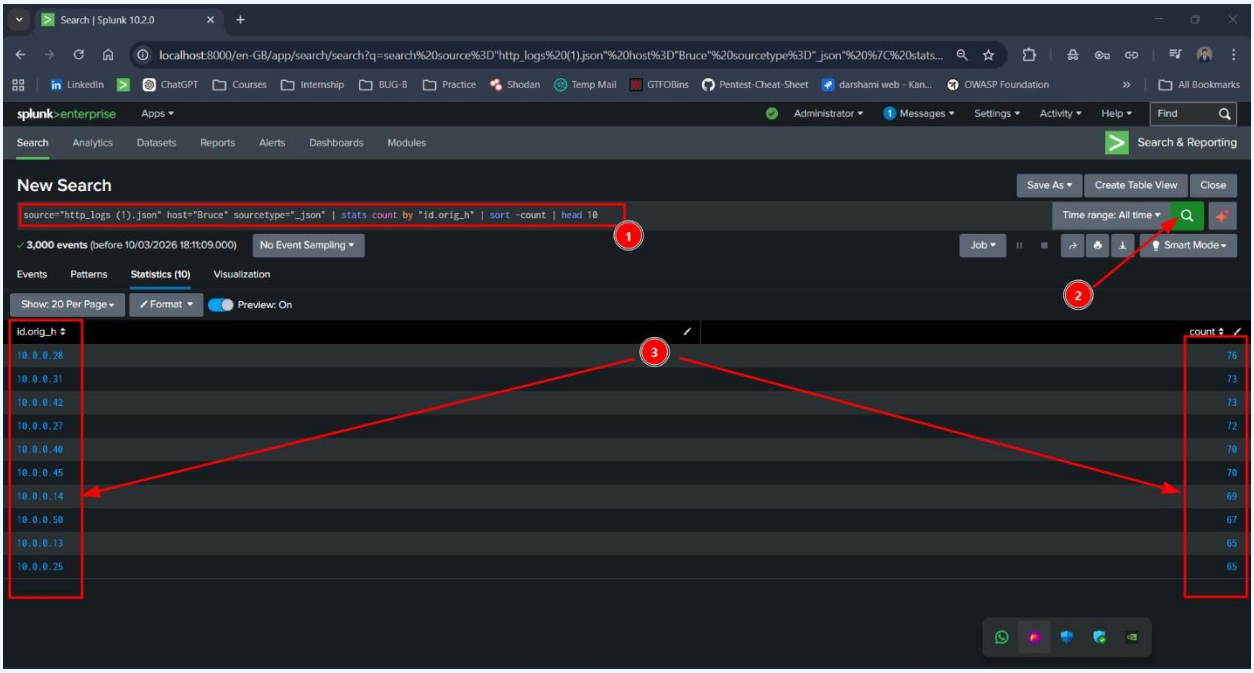
### Analysis Result

This query helps in identifying:

- IP addresses generating high web traffic
- Systems that may be performing automated requests
- Possible scanning or abnormal activity from specific hosts




### Task 2: Server Error Analysis (HTTP 5xx)

HTTP 5xx status codes indicate errors from the server side. These occur when the server fails to process a request properly.

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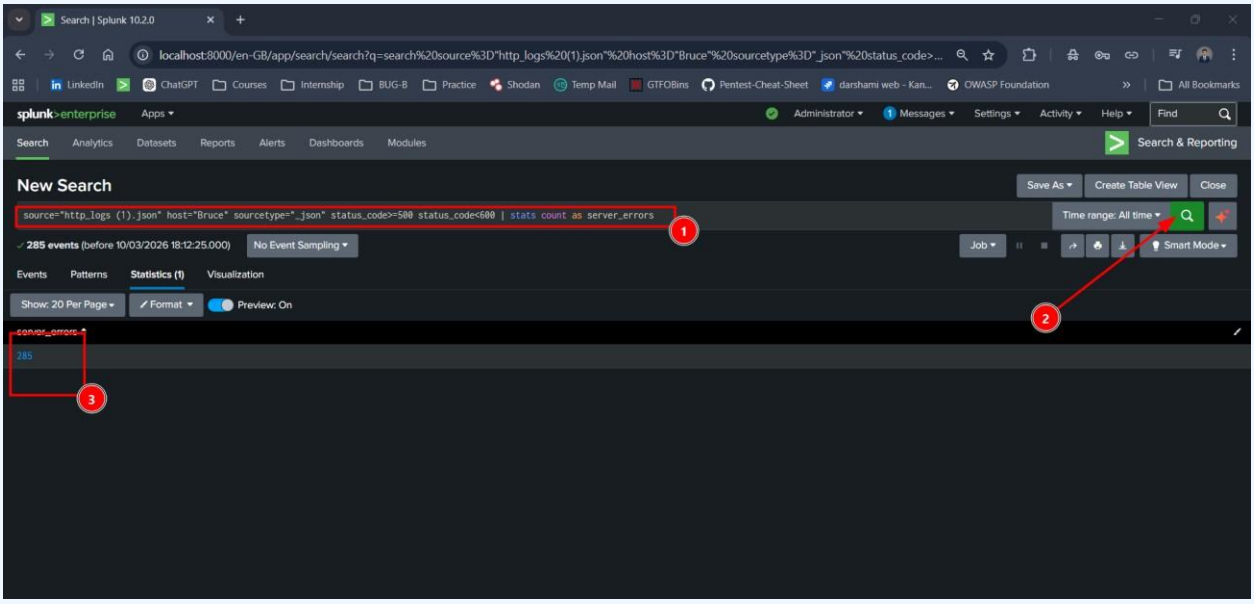
**SPL Query**

```
index=http_lab sourcetype="json" status_code>=500 status_code<600
| stats count as server_errors
```

**Query Explanation**

- **status_code>=500 status_code<600**
  Filters all server error responses (5xx range).
- **stats count**
  Counts the total number of matching events.
- **as server_errors**
  Renames the output field for clarity.




**Analysis Result**

This query gives the total number of server-side errors found in the logs.

A high number of 5xx errors may indicate:

- Issues in the web application
- Server overload or misconfiguration

**Task 3: User-Agent Analysis**

User-agent values provide information about the client or tool that is making the HTTP request. This helps in understanding whether the traffic is coming from a normal browser or from an automated script or tool.

In many cases, attackers and scanning tools use specific user-agent strings, which can be identified during log analysis.

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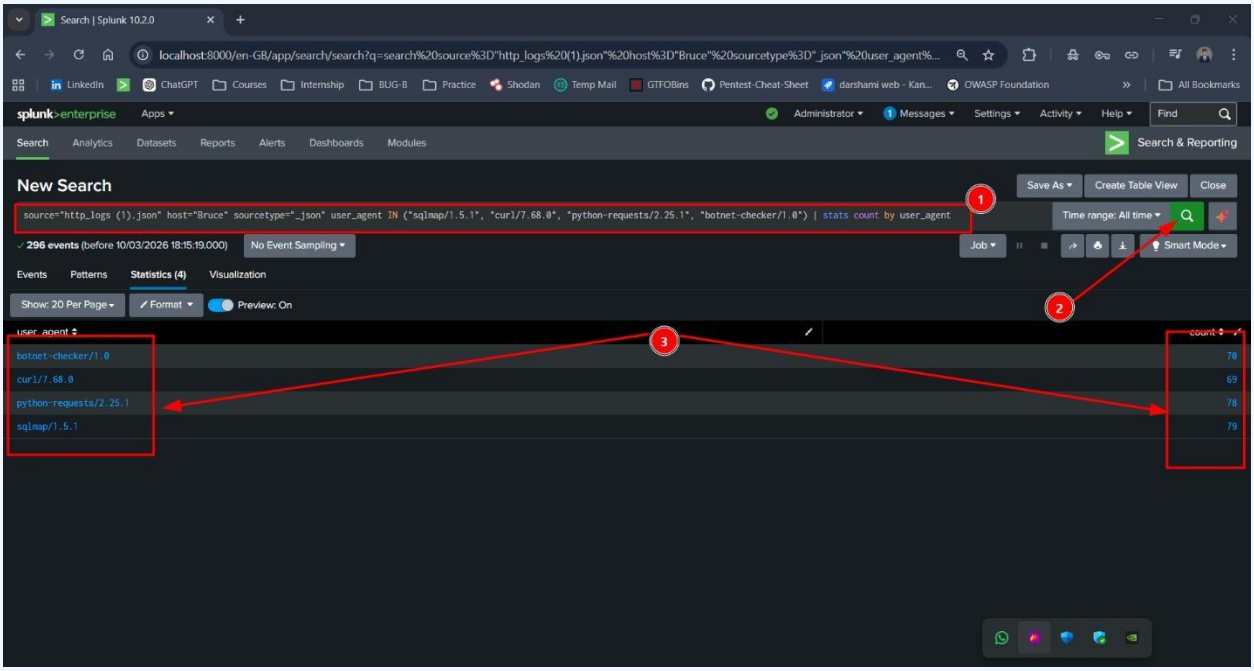
### SPL Query

```
index=http_lab sourcetype="json"
user_agent IN ("sqlmap/1.5.1","curl/7.68.0","python-requests/2.25.1","botnet-checker/1.0")
| stats count by user_agent
```

### Query Explanation

- **user_agent IN (...)**
  Filters the logs to show only requests coming from selected user-agent values.
- **stats count by user_agent**
  Counts how many times each user-agent appears in the logs.




### Analysis Result

The results show multiple requests generated by tools rather than standard web browsers.

Some commonly identified user-agents include:

- **slap** – used for testing SQL injection vulnerabilities
- **curl** – command-line tool often used for sending requests manually
- **python-requests** – commonly used in scripts and automation
- **botnet-checker** – may indicate automated or bot-related activity

The presence of these user-agents suggests that some of the traffic may not be normal user activity.

**Observation**

- These tools are often used for testing, scanning, or automation
- Repeated requests from such user-agents may indicate scripted activity
- It helps in distinguishing between human users and automated traffic

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**Task 4: Data Transfer Analysis**

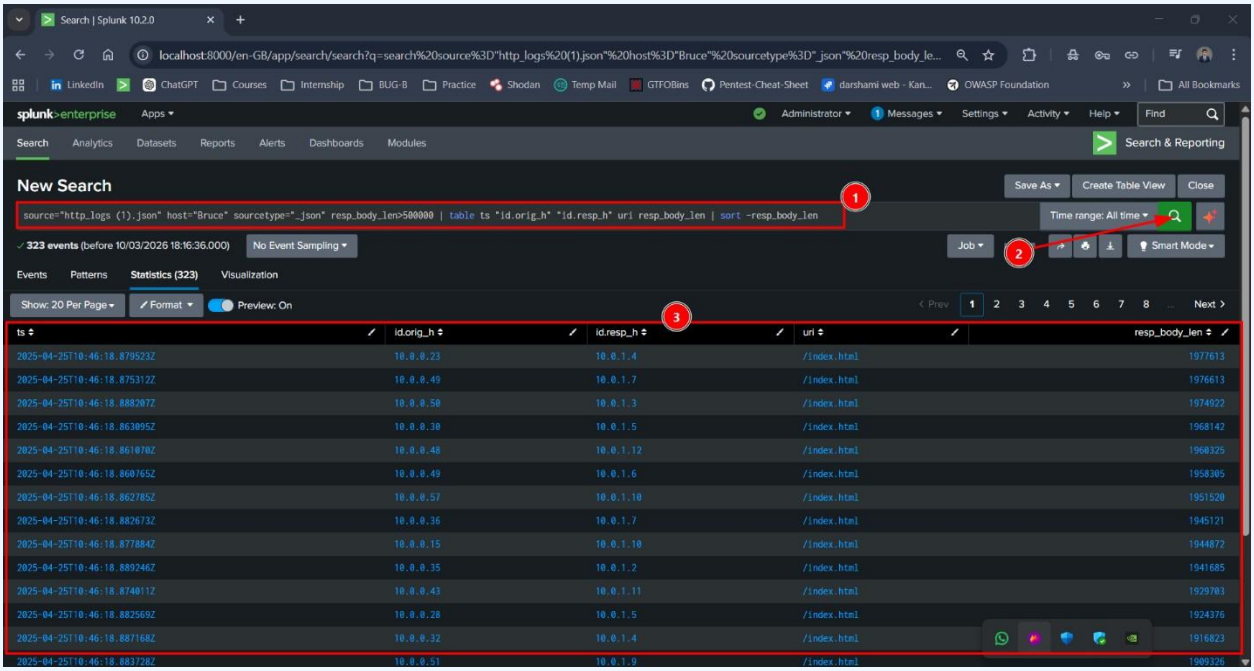
Large response sizes in HTTP logs can indicate file downloads, uploads, or transfer of data between client and server. Monitoring such activity is important to understand how data is being exchanged over the network.

**SPL Query**

```
index=http_lab sourcetype="json" resp_body_len>500000
| table ts "id.orig_h" "id.resp_h" uri resp_body_len
| sort -resp_body_len
```

**Query Explanation**

- **resp_body_len>500000**
  Filters all HTTP responses where the size is greater than 500 KB. This helps in focusing only on large data transfers.
- **table**
  Displays important fields such as timestamp, source IP, destination IP, requested resource, and response size.
- **sort -resp_body_len**
  Sorts the results in descending order to show the largest responses at the top.




### Analysis Result

The output of this query shows HTTP requests where large amounts of data were transferred.

From the results, it can be observed that:

- Some responses have significantly large sizes
- These requests may involve downloading files or accessing large resources
- Certain IP addresses are involved in repeated large transfers

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### Observation

- Large data transfers are common in normal usage but should be reviewed carefully
- If large responses occur frequently from the same IP, it may indicate unusual behavior
- Monitoring such activity helps in identifying how data is flowing within the system

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### Security Insights

From the overall HTTP log analysis, different types of activities can be identified which are useful for basic security monitoring:

1. **Web Scanning Activity**
   A high number of requests from specific IP addresses may indicate scanning or enumeration.

2. **Testing of Web Applications**
   Tools like sqlmap are often used to check for vulnerabilities such as SQL injection.
3. **Automated Traffic**
   Some requests are generated by scripts or tools instead of normal browsers.
4. **Data Transfer Activity**
   Large response sizes may indicate file downloads or data movement across the network.

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**Overall Understanding**

This analysis shows how HTTP logs can be used to understand user activity, detect unusual patterns, and monitor basic security events.

Splunk makes it easier to perform such analysis by providing quick search and filtering capabilities.

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## Conclusion

In this lab, HTTP logs were successfully uploaded and analyzed using Splunk. The process helped in understanding how JSON formatted log data is handled and how queries can be used to examine web traffic.

During the analysis, it was possible to identify:

- IP addresses generating high web traffic
- Server-side errors (5xx responses)
- Requests from automated tools and scripts
- Large data transfers in HTTP responses

This lab provided a basic understanding of how log analysis can be used to monitor web activity and detect unusual patterns.
These techniques are useful in areas such as security monitoring, intrusion detection, and incident analysis.